

ClimateScope: An Interactive Visual Intelligence Framework for Global Weather Trends and Risk Quantification

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Abstract—Climate change has intensified the frequency and severity of extreme weather events, making it increasingly important to transform complex meteorological data into actionable insights. This paper presents *ClimateScope*, an interactive visual intelligence framework for analyzing global weather trends and quantifying environmental risk. By integrating temperature, precipitation, and wind data, the proposed system introduces a Composite Climate Risk (CCR) model to assess regional severity. The system leverages a Streamlit-based architecture for real-time data exploration and visualization. Experimental results indicate a consistent rise in global temperature along with a disproportionately higher increase in climate risk, emphasizing the need for intelligent monitoring and decision-support systems.

Index Terms—Climate Dashboard, Risk Analysis, Streamlit, Weather Analytics, Data Visualization.

I. INTRODUCTION

Climate change has emerged as a critical global challenge, significantly increasing the frequency, intensity, and unpredictability of extreme weather events. Although large-scale meteorological datasets are readily available, extracting meaningful insights from such data remains a complex task due to its volume and variability.

To address this challenge, this paper introduces *ClimateScope*, an interactive visual intelligence framework that combines data analytics with visualization techniques to assess environmental risk. Unlike traditional dashboards, the proposed system integrates a composite risk scoring mechanism, enabling users to better understand climate patterns and associated risks.

Key Contributions:

- Development of a Composite Climate Risk (CCR) model integrating temperature, precipitation, and wind
- Design of an interactive dashboard for real-time climate data exploration
- Identification of accelerating climate risk trends using visual analytics

II. PROPOSED METHODOLOGY

A. Data Acquisition and Pre-processing

The dataset is obtained from the Global Weather Repository, containing multi-dimensional environmental data. Preprocessing steps include handling missing values, removing inconsistencies, and normalizing variables to ensure comparability across different scales.

B. Mathematical Framework for Risk Analysis

The Composite Climate Risk score is calculated as:

$$R_{score} = 0.45(T_{norm}) + 0.25(P_{norm}) + 0.30(W_{norm}) \quad (1)$$

Weight Justification: Temperature has the highest weight due to its long-term impact, while wind and precipitation contribute to extreme events and localized variability.

C. Heuristic Criteria for Extreme Events

$$E = \begin{cases} \text{Heatwave} & \text{if } T > 40^{\circ}\text{C} \\ \text{Flash Flood} & \text{if } P > 100 \text{ mm} \\ \text{Storm} & \text{if } W > 60 \text{ kph} \end{cases} \quad (2)$$

These threshold-based conditions provide a simple mechanism for identifying extreme weather events.

III. SYSTEM ARCHITECTURE AND IMPLEMENTATION

The ClimateScope system is implemented using a Python backend integrated with a Streamlit frontend. The system enables real-time interaction, dynamic filtering, and visualization of climate data.

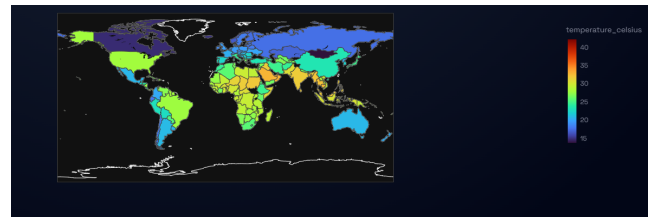


Fig. 1. Geospatial distribution of global temperature anomalies

IV. RESULTS AND DISCUSSION

A. Quantitative Trend Analysis

The analysis indicates that global temperature has increased by approximately 1.2%, while the Composite Climate Risk Index has increased by 2.1%. This highlights a compounding effect where multiple environmental factors collectively amplify overall climate risk.

TABLE I
CLIMATE ANOMALY SUMMARY (2025-2026)

Metric	Mean	Max	Trend
Temperature (°C)	24.5	48.2	+1.2%
Precipitation (mm)	12.4	145.0	+0.8%
Risk Index	0.42	0.89	+2.1%

B. Risk Score Distribution

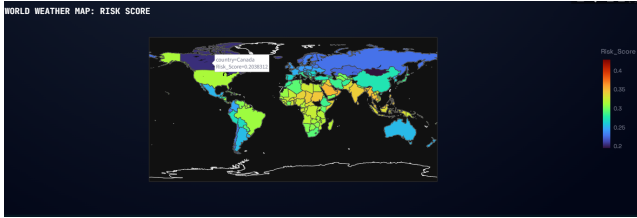


Fig. 2. Global distribution of Composite Climate Risk (CCR) scores

The results show that climate risk is unevenly distributed across regions, with certain areas experiencing significantly higher risk levels.

C. Temporal Risk Trends

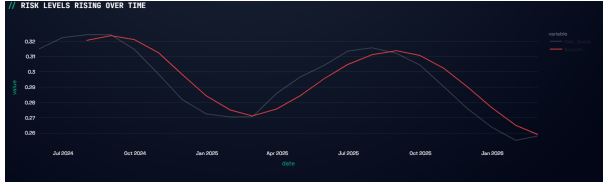


Fig. 3. Temporal variation of the CCR score

Temporal analysis reveals periodic fluctuations influenced by seasonal patterns, along with an overall increase in variability.

D. Seasonal Heatmap and Variation



Fig. 4. Monthly temperature variations showing extended seasonal intensity

The heatmap indicates that temperature intensity is extending beyond traditional seasonal boundaries.

E. Interpretation of Extreme Events



Fig. 5. Cumulative growth of extreme weather events

Extreme events exhibit nonlinear growth, suggesting accelerating climate instability.

F. Real-World Applications

- Disaster management and early warning systems
- Agricultural planning and crop risk assessment
- Climate policy development and environmental monitoring

G. Limitations

The model relies on static thresholds and does not incorporate real-time data streams, which may limit predictive accuracy.

V. CONCLUSION

This paper presented ClimateScope, an interactive framework for analyzing global weather trends and quantifying climate risk. The findings demonstrate that climate risk is increasing at a faster rate than individual environmental parameters.

The study highlights the importance of integrating visualization with analytical models to support informed decision-making. Future work will focus on incorporating real-time data and advanced predictive techniques.

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